

Department of Computer Science and Engineering

Assignment Questions

Course Code & Name : CSA0702 – COMPUTER NETWORKS

Assignment Title:	Design a VLAN-based enterprise LAN containing separate VLANs for Administration, Faculty, Students, and Guests. Configure VLANs and verify communication.
Course Outcome(s) – CO:	CO1: Identify and explain fundamental networking concepts, network devices, topologies, protocols, and layered network architectures. CO2: Configure IP addressing, subnetting, switching, and routing techniques for establishing computer networks using Cisco Packet Tracer. CO3: Analyze network traffic, addressing information, VLAN communication, and connectivity using appropriate network simulation and packet-analysis tools. CO4: Evaluate network performance, connectivity, segmentation, and security requirements for different enterprise networking scenarios. CO5: Design and implement VLAN-based enterprise networks by creating separate VLANs for Administration, Faculty, Students, and Guests, configuring inter-VLAN communication, and verifying connectivity using Cisco Packet Tracer.
Bloom's Taxonomy Level	L6 –Design; L5 – Evaluate; L4- Analyze; L3-Configure; L2- Explain
SDG Mapping (SDG 1-17)	SDG 4-Quality Education-Develops practical networking knowledge, digital skills, analytical abilities, and hands-on technical competency. SDG 9-Industry, Innovation and Infrastructure-Promotes the design and implementation of reliable, secure, scalable, and innovative ICT/network infrastructure. SDG 11-Sustainable Cities and Communities-
	Provides networking skills relevant to connected campuses, smart buildings, smart cities, and resilient digital services.

Problem Statement: Design and implement a VLAN-based enterprise Local Area Network (LAN) using Cisco Packet Tracer for an organization with four departments: Administration, Faculty, Students, and Guests. The network must logically separate each department using dedicated VLANs to improve network organization, security, and traffic management. Configure appropriate VLANs, assign switch ports to the respective VLANs, configure IP addressing and inter-VLAN communication, and verify connectivity between authorized network segments. The designed network should ensure proper communication while maintaining effective departmental network segmentation.

Assessment Rubrics

Criteria (CO Mapping)	Max Marks	Excellent	Good	Satisfactory	Needs Improvement
Network Design & Topology	10	Logical and complete topology	Minor design issues	Basic topology	Incomplete/incorrect topology
VLAN Configuration	15	All VLANs correctly created and configured	Minor errors	Most VLANs configured	Major configuration errors
IP Addressing & Port Assignment	10	Correct addressing and port allocation	Few errors	Partially correct	Incorrect configuration
Inter-VLAN Communication	15	Fully configured and working	Minor issues	Partially working	Not working
Connectivity Verification	15	Successfully verifies all required connections	Minor verification issues	Basic verification	Unable to verify
Troubleshooting & Analysis	25	Identifies and resolves issues independently	Resolves minor issues	Needs some guidance	Unable to troubleshoot

Reflection: Design Decisions, SDG Relevance & Learning Outcomes	10	Clearly connects the activity with SDG 9 and explains its relevance to secure and efficient digital infrastructure.	Provides a clear and relevant reflection.	Provides basic explanation of some design decisions.	Unable to relate the activity to an SDG.
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1. Aim

To plan, configure and validate a segmented Virtual LAN (VLAN) architecture on Cisco Packet Tracer for a campus network with four functional zones — Administration, Faculty, Students and Guests — by allocating access ports on a Layer-2 switch, building an IEEE 802.1Q trunk between the switch and a router, enabling Router-on-a-Stick (ROAS) inter-VLAN routing, and confirming both same-VLAN and cross-VLAN reachability.

2. Algorithm

- Step 1 – Addressing Plan: Reserve four VLAN identifiers (VLAN 15: Administration, VLAN 25: Faculty, VLAN 35: Students, VLAN 45: Guests) and map each to a private /24 block drawn from the 172.16.0.0 range, with the first usable host address on each subnet reserved for the gateway.
- Step 2 – Physical Layout: Drag one Cisco 2911 Integrated Services Router, one Cisco Catalyst 2950-24TT switch and four end-user PCs onto the canvas; wire each PC to a switch access port and link the switch to the router with a Copper Straight-Through cable.
- Step 3 – VLAN Database: On the switch, enter global configuration mode and create four VLAN entries (15, 25, 35, 45), naming each after its department.
- Step 4 – Access Port Mapping: Place each departmental port block into static access mode and bind it to the matching VLAN using switchport mode access and switchport access vlan <id>.
- Step 5 – Trunk Setup: Convert the switch-to-router uplink into an 802.1Q trunk with switchport mode trunk so that frames from all four VLANs can share the single physical link.
- Step 6 – Router-on-a-Stick: Bring up the router's physical interface with no shutdown (leaving it without an IP), then carve out four logical subinterfaces, tag each with dot1Q encapsulation for its VLAN, and assign the corresponding gateway address.
- Step 7 – Endpoint Configuration: On every PC, set a static IP address, a /24 mask and the default gateway that matches its department's subnetwork.
- Step 8 – Validation: Run show vlan brief, show interfaces trunk and show ip route on the network devices, then use ICMP ping tests to confirm both intra-VLAN and inter-VLAN paths are functioning.

3. Theory

3.1 Virtual LANs

A switch that has not been segmented behaves as one flat broadcast domain — every ARP request or broadcast frame reaches every connected port. A VLAN carves that single physical switch into several independent logical broadcast domains, so traffic from one department stays isolated from another even though the devices share the same hardware. This lowers unnecessary broadcast traffic, contains faults to a smaller zone, and lets an administrator group users by function rather than by which cable they happen to plug into.

3.2 802.1Q Trunking

When multiple VLANs need to travel across a single cable — for example between a switch and a router — the link is configured as a trunk. The IEEE 802.1Q standard tags each Ethernet frame with a small 4-byte field carrying a 12-bit VLAN ID before it leaves the sending device. The receiving side reads this tag to work out which VLAN the frame belongs to, which is what allows one physical trunk port to carry traffic for many logical networks at once.

3.3 Router-on-a-Stick Inter-VLAN Routing

Switches forward frames using Layer-2 addressing only, so two hosts sitting in different VLANs cannot reach each other without help from a Layer-3 device. The Router-on-a-Stick model solves this with a single router interface connected to the switch's trunk port; that one physical interface is then split into several virtual subinterfaces, each tied to one VLAN tag and configured with the IP address that acts as the default gateway for that department.

3.4 Alignment with SDG 9

Because ROAS lets several logical networks share one router interface and one switch, the organisation avoids buying a dedicated router and switch for every department. This cuts hardware and power costs while still giving each zone its own secured, manageable segment — supporting the SDG 9 goal of building resilient, efficient and innovative digital infrastructure for institutions and smart campuses.

4. Network Topology

The design follows a two-tier star topology built around a Router-on-a-Stick core. A Cisco Catalyst 2950-24TT access switch terminates four department workstations on isolated access ports, and a single 802.1Q trunk carries all VLAN traffic up to a Cisco 2911 router that performs inter-VLAN routing on logical subinterfaces.

```

+-----+
| Router1 (Cisco 2911)
|
| Physical Interface: GigabitEthernet0/1 (no IP address assigned)
|
| Gig0/1.15: 172.16.15.1/24 (Admin GW) | Gig0/1.25: 172.16.25.1/24 (Faculty
GW) |
| Gig0/1.35: 172.16.35.1/24 (Student GW) | Gig0/1.45: 172.16.45.1/24 (Guest GW)
|
+-----+-----+
+-----+
802.1Q Trunk Link (Switch Gig0/2 <-> Router Gig0/1)
+-----+
| Switch1 (Cisco Catalyst 2950-24TT)
|
+-----+-----+-----+-----+
+-----+
Fa0/2          Fa0/7          Fa0/12          Fa0/17
(VLAN 15)      (VLAN 25)      (VLAN 35)      (VLAN 45)
|              |              |              |
v              v              v              v
PC-Admin      PC-Faculty      PC-Student      PC-Guest
192.168.10.10  192.168.20.10   192.168.30.10   192.168.40.10
GW: 192.168.10.1  GW: 192.168.20.1   GW: 192.168.30.1  GW: 192.168.40.1

```

4.1 Device & Link Summary

Device	Model / Type	Interface	Connects To	Role
Router1	Cisco 2911	Gig0/1 (.15/.25/.35/.45)	Switch1 Gig0/2	Trunk / Inter-VLAN gateway
Switch1	Catalyst 2950-24TT	Gig0/2	Router1 (Gig0/1)	802.1Q trunk uplink
Switch1	Catalyst 2950-24TT	Fa0/2	PC-Admin	Access (VLAN 15)
Switch1	Catalyst 2950-24TT	Fa0/7	PC-Faculty	Access (VLAN 25)
Switch1	Catalyst 2950-24TT	Fa0/12	PC-Student	Access (VLAN 35)
Switch1	Catalyst 2950-24TT	Fa0/17	PC-Guest	Access (VLAN 45)

5. IP Addressing and VLAN Details

Table 5.1 — VLAN Allocation & Port Assignment

VLAN ID	Name	Department	Port Range	Used Port	Network	Mask
15	Administration	Admin & Mgmt	Fa0/2 – Fa0/6	Fa0/2	192.168.10.0/24	255.255.255.0
25	Faculty	Faculty & Staff	Fa0/7 – Fa0/11	Fa0/7	192.168.20.0/24	255.255.255.0
35	Students	Student Body	Fa0/12 – Fa0/16	Fa0/12	192.168.30.0/24	255.255.255.0
45	Guests	Guest Access	Fa0/17 – Fa0/21	Fa0/17	192.168.40.0/24	255.255.255.0
Trunk	All (15–45)	Uplink	Gig0/2	Gig0/2	802.1Q Tagged	N/A

Table 5.2 — Host & Subinterface Addressing

Device	Interface	VLAN	IP Address	Mask	Gateway
PC-Admin	FastEthernet0	15	192.168.10.10	255.255.255.0	192.168.10.1
PC-Faculty	FastEthernet0	25	192.168.20.10	255.255.255.0	192.168.20.1
PC-Student	FastEthernet0	35	192.168.30.10	255.255.255.0	192.168.30.1
PC-Guest	FastEthernet0	45	192.168.40.10	255.255.255.0	192.168.40.1
Router1	Gig0/1.15	15	192.168.10.1	255.255.255.0	N/A (ROAS GW)
Router1	Gig0/1.25	25	192.168.20.1	255.255.255.0	N/A (ROAS GW)
Router1	Gig0/1.35	35	192.168.30.1	255.255.255.0	N/A (ROAS GW)
Router1	Gig0/1.45	45	192.168.40.1	255.255.255.0	N/A (ROAS GW)

6. Configuration

6.1 Switch1 CLI

```
Switch(config-if)#exit
Switch(config)#interface fa0/16
Switch(config-if)#switchport mode access
Switch(config-if)#switch access vlan 40
Switch(config-if)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Switch#show vlan brief

VLAN Name                Status    Ports
----
1    default              active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                           Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                           Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                           Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                           Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                           Gig0/1, Gig0/2
10   Administration        active    Fa0/1
20   faculty               active    Fa0/6
30   students              active    Fa0/11
40   guests                active    Fa0/16
1002 fddi-default          active
1003 token-ring-default   active
1004 fddinet-default      active
1005 trnet-default        active
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#interface gigabitEthernet0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#no shutdown
Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
```

6.2 Router1 ROAS CLI

```
outer#show interfaces trunk

outer#enable
outer#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
outer(config)#interface GigabitEthernet0/0
outer(config-if)#no shutdown
outer(config-if)#exit
outer(config)#end
LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

outer#
SYS-5-CONFIG_I: Configured from console by console
configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
outer(config)#
outer(config)#enable
Incomplete command.
outer(config)#configure terminal
Invalid input detected at '^' marker.
outer(config)#interface gigabitEthernet0/0
outer(config-if)#no shutdown
outer(config-if)#exit
outer(config)#end
outer#
SYS-5-CONFIG_I: Configured from console by console

outer#show interfaces status
Invalid input detected at '^' marker.

outer#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
outer(config)#interface gigabitEthernet0/0.10
outer(config-subif)#encapsulation dot1Q 10
outer(config-subif)#ip address 192.168.10.1 255.255.255.0
outer(config-subif)#exit
LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down
```

7. Verification

7.1 Switch VLAN & Trunk Status

```
Switch#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Switch#show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/2, Fa0/3, Fa0/4, Fa0/5 Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10 Administration	active	Fa0/1
20 faculty	active	Fa0/6
30 students	active	Fa0/11
40 guests	active	Fa0/16
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```
Switch#enable
```

7.2 Router Interface & Routing Table

```
Switch#enable
Switch#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/1	unassigned	YES	manual	up	up
FastEthernet0/2	unassigned	YES	manual	up	up
FastEthernet0/3	unassigned	YES	manual	up	up
FastEthernet0/4	unassigned	YES	manual	up	up
FastEthernet0/5	unassigned	YES	manual	up	up
FastEthernet0/6	unassigned	YES	manual	down	down
FastEthernet0/7	unassigned	YES	manual	down	down
FastEthernet0/8	unassigned	YES	manual	down	down
FastEthernet0/9	unassigned	YES	manual	down	down
FastEthernet0/10	unassigned	YES	manual	down	down
FastEthernet0/11	unassigned	YES	manual	down	down
FastEthernet0/12	unassigned	YES	manual	down	down
FastEthernet0/13	unassigned	YES	manual	down	down
FastEthernet0/14	unassigned	YES	manual	down	down
FastEthernet0/15	unassigned	YES	manual	down	down
FastEthernet0/16	unassigned	YES	manual	down	down
FastEthernet0/17	unassigned	YES	manual	down	down
FastEthernet0/18	unassigned	YES	manual	down	down
FastEthernet0/19	unassigned	YES	manual	down	down
FastEthernet0/20	unassigned	YES	manual	down	down
FastEthernet0/21	unassigned	YES	manual	down	down

```
--More--
```

Router1# show

```

router#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 unassigned YES unset up up
GigabitEthernet0/0.10 192.168.10.1 YES manual up up
GigabitEthernet0/0.20 192.168.20.1 YES manual up up
GigabitEthernet0/0.30 192.168.30.1 YES manual up up
GigabitEthernet0/0.40 192.168.40.1 YES manual up up
GigabitEthernet0/1 unassigned YES unset down down
GigabitEthernet0/2 unassigned YES unset administratively down down
lan1 unassigned YES unset administratively down down

router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
  192.168.10.0/24 is directly connected, GigabitEthernet0/0.10
  192.168.10.1/32 is directly connected, GigabitEthernet0/0.10
192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
  192.168.20.0/24 is directly connected, GigabitEthernet0/0.20
  192.168.20.1/32 is directly connected, GigabitEthernet0/0.20
192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
  192.168.30.0/24 is directly connected, GigabitEthernet0/0.30
  192.168.30.1/32 is directly connected, GigabitEthernet0/0.30
192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
  192.168.40.0/24 is directly connected, GigabitEthernet0/0.40
  192.168.40.1/32 is directly connected, GigabitEthernet0/0.40

```

7.3 ICMP Connectivity Results

Source	Destination	Traffic Type	Result	Remarks
PC-Admin (.15.20)	Gateway 192.168.10.10	Intra-VLAN	4/4 success	Sub-1ms, local subinterface reachable
PC-Admin (.15.20)	PC-Faculty (.20.10)	Inter-VLAN 15→25	4/4 success	Routed via ROAS subinterfaces
PC-Faculty (.25.20)	PC-Student (.30.10)	Inter-VLAN 25→35	4/4 success	Full bidirectional reachability
PC-Student (.35.20)	PC-Guest (.40.10)	Inter-VLAN 35→45	4/4 success	Full bidirectional reachability

8. Troubleshooting

Symptom	Likely Cause	Diagnostic Command	Fix
Host cannot reach its own gateway	Port bound to wrong VLAN, or host IP outside its subnet	show vlan brief / ipconfig	Re-apply switchport access vlan <id>; correct static IP
Intra-VLAN works, inter-VLAN pings time out	Uplink not trunked, or router subinterface parent is down	show interfaces trunk / show ip int brief	Set switchport mode trunk; no shutdown on Gig0/1
Traffic lands on the wrong subnet or is dropped	Subinterface dot1Q tag doesn't match switch VLAN ID	show running-config interface Gig0/1.x	Correct encapsulation dot1Q <id> to match
Local subnet reachable, nothing beyond it	Default gateway missing or wrong on host	ipconfig /all	Set gateway to the matching router subinterface IP

9. Result

- Four VLANs — 10 (Administration), 20 (Faculty), 30 (Students) and 40 (Guests) — were created on Switch1 with dedicated static access port ranges for each department.
- An IEEE 802.1Q trunk was brought up on Switch1 Gig0/2, carrying tagged traffic for all four VLANs to Router1.
- Router-on-a-Stick inter-VLAN routing was configured across subinterfaces Gig0/1.15 through Gig0/1.45, each carrying the correct dot1Q tag and gateway address.

ICMP testing confirmed 100% success for both intra-VLAN and inter-VLAN traffic, verifying that the four departments remain logically segmented while still able to communicate through the router — in line with UN SDG 9 .